Database Design Document

University Internship & Industry Matching Portal

1. Introduction

1.1 Purpose

This document defines the database architecture for the University Internship & Industry Matching Portal. It provides a detailed design of tables, relationships, and constraints required to store and manage all system data.

1.2 Scope

The database will store all information for students, companies, administrators, internships, projects, competitions, applications, and document storage references. It will be implemented using MySQL and deployed on Azure Database Service.

2. Entities and Relationships

The system database includes the following main entities:

2.1 User

Stores all registered users.

- user_id (PK)
- full_name
- email (unique)
- password_hash
- role (Student, Company, Admin)
- contact_number
- created_at

- updated_at

2.2 Student Profile

Stores student-specific information.

- student_id (PK, FK to user_id)
- university_id
- department
- resume_link (Azure Blob Storage URL)
- skills
- GPA
- created_at
- updated_at

2.3 Company Profile

Stores company-specific information.

- company_id (PK, FK to user_id)
- company_name
- industry
- address
- description
- website_url
- created_at
- updated_at

2.4 Internship

Stores internship postings.

- internship_id (PK)
- company_id (FK)

- title
- description
- requirements
- duration
- location
- status (Active, Closed)
- created_at
- updated_at

2.5 Internship Application

Tracks student applications for internships.

- application_id (PK)
- internship_id (FK)
- student_id (FK)
- status (Applied, Shortlisted, Interview, Selected, Rejected)
- applied_at
- updated_at

2.6 Project

Stores university projects and startups.

- project_id (PK)
- title
- description
- category (Research, Startup, Product)
- leader_id (FK to user_id)
- required_skills
- team_size

- start_date
- end_date
- status (Active, Completed)
- created_at
- updated_at

2.7 Project Participation

Tracks students participating in projects.

- participation_id (PK)
- project_id (FK)
- student_id (FK)
- role (Developer, Researcher, Designer, etc.)
- status (Pending, Accepted, Rejected)
- joined_at
- updated_at

2.8 Competition

Stores competitions.

- competition_id (PK)
- title
- description
- organizer (FK to user_id or external organizer)
- category
- eligibility_criteria
- start_date
- end_date
- registration_link

- created_at
- updated_at

2.9 Competition Participation

Tracks student participation in competitions.

- competition_participation_id (PK)
- competition_id (FK)
- student_id (FK)
- status (Registered, Submitted, Won, Participated)
- applied_at
- updated_at

2.10 Document Storage

Stores references to uploaded documents.

- document_id (PK)
- student_id (FK)
- type (Resume, Certificate, Project File)
- storage_url (Azure Blob URL)
- uploaded_at
- updated_at

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3. Relationships

- **User** → **Student Profile / Company Profile**: One-to-one
- **Company** → **Internship**: One-to-many
- **Student** → **Internship Application**: One-to-many
- **Project** → **Project Participation** → **Student**: Many-to-many
- **Competition** → **Competition Participation** → **Student**: Many-to-many

- **Student** → **Document Storage**: One-to-many

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4. ER Diagram

A visual ER Diagram should be created including:

- Entities: User, Student Profile, Company Profile, Internship, Internship Application, Project, Project Participation, Competition, Competition Participation, Document Storage
- Relationships with cardinalities
- Primary and Foreign keys

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5. Constraints and Indexes

- Unique email for users
- Foreign key constraints for referential integrity
- Indexes on frequently queried fields: student_id, company_id, internship_id, project_id, competition_id

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6. Normalization

- Database is normalized to **3NF**
- Avoid data redundancy
- Ensure efficient queries

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7. Data Retention and Security

- Resume and documents stored in Azure Blob with secure access
- Student, company, and admin data stored securely with encryption at rest
- Regular backups via Azure Database Service

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8. Summary

The database design ensures data integrity, security, scalability, and supports all system functionalities, including internships, projects, competitions, and document storage.